

The Privacy Externality Game: Strategic Budget Choices in Sampling-Based Individual Differential Privacy

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Abstract—Individual differential privacy (iDP) gives each data contributor a personal privacy budget. The currently dominant realisation—sampling-based iDP—turns each contributor’s chosen budget into a per-sample inclusion rate, with a single noise multiplier shared across the dataset. We show that this realisation creates a privacy externality: lowering one’s own budget mechanically raises the membership-inference (MIA) advantage on every other data point. Although every individual (ε_i, δ) -iDP guarantee is honoured, the realised privacy of any one contributor is determined by the choices of all others. We model the resulting strategic situation as a finite, normal-form game, the Privacy Externality Game (PEG), and study its equilibria. We prove that the privacy externality is monotone: $\partial A_i / \partial \varepsilon_j \leq 0$ for $j \neq i$ under the mechanism’s binding constraints. We then characterise the game under two realistic awareness regimes—naïve (players perceive only their nominal ε_i) and aware (players perceive the realised MIA advantage)—and show that the aware regime exhibits no pure Nash equilibrium in canonical settings, admitting only mixed equilibria. A coalition Stackelberg analysis recovers the attack of Kaiser et al. (SaTML 2026) as the rational best response of any coordinated set of data holders. The implication is that the empirical 62%-vulnerability attack rate observed in iDP is not adversarial but equilibrium behaviour: even non-malicious players in iDP impose privacy externalities on every other contributor.

Index Terms—Differential privacy, individual differential privacy, game theory, Nash equilibrium, collusion, membership inference.

I. INTRODUCTION

Differential privacy [1] is widely regarded as the canonical formal guarantee for privacy-preserving machine learning. In its classical form, a single privacy budget (ε, δ) is shared by every record in the database, and every record is protected to exactly the same extent. Yet privacy preferences are heterogeneous: a clinical patient and a casual user contributing data to the same model often place very different values on their privacy. Recognising this, *individual differential privacy* (iDP) replaces the single (ε, δ) guarantee with a per-record contract (ε_i, δ) , and a number of mechanisms have been proposed to deliver such per-record contracts in deep learning [2]–[4]. The most widely-adopted realisation, sampling-based iDP introduced by Boenisch et al. [2], achieves heterogeneous budgets by calibrating each record’s Poisson inclusion probability p_i against a single shared noise multiplier σ so that, for fixed batch size and iteration count, every group of records satisfies its own (ε_g, δ) -DP bound.

We focus on a fundamental and overlooked feature of this realisation. *Because every record shares the same noise*

multiplier and a common expected batch size, each record’s sampling rate is mechanically coupled to every other record’s chosen budget. Concretely, lowering one’s own ε_i forces other contributors’ sampling rates upward to keep the expected batch size fixed—which, as recent empirical work has shown [5], inflates their membership-inference (MIA) advantage even though every formal (ε_g, δ) -iDP guarantee continues to hold. The promise of individual control is therefore broken in a structural sense: an individual’s *actual* privacy risk depends not on her own choice but on the joint choice of the entire population.

a) *Our viewpoint: this is a game.*: The natural framing for any choice problem whose outcome is determined by the choices of multiple self-interested parties is game theory. Pejó et al. [6] introduced a two-player game on the privacy-accuracy plane (the *Price of Privacy*), but they were studying a mechanism in which a player’s privacy choice affected only her *own* accuracy. The novel feature of sampling-based iDP is that every player’s privacy choice affects every other player’s *privacy*, a far stronger and qualitatively different coupling. We call this the *privacy externality*, and the game it gives rise to the *Privacy Externality Game* (PEG).

b) *Contributions.*: This paper makes the following contributions.

- 1) **Formalisation.** We define the n -player Privacy Externality Game (PEG) in which each player picks a per-group budget ε_i and the sampling-based iDP mechanism then determines the noise multiplier σ and the per-group sampling rates p_g . We split player utility into two awareness regimes that capture realistic contracting situations: *naïve* players, who price privacy via their nominal ε_i , and *aware* players, who price privacy via the realised MIA advantage $A_i(\varepsilon)$.
- 2) **Externality lemma.** We prove that the externality is monotone: under the binding constraints of the mechanism, $\partial A_i / \partial \varepsilon_j \leq 0$ for every $j \neq i$. Lowering any other player’s budget never reduces my realised vulnerability.
- 3) **Equilibrium characterisation.** For the aware regime, we observe that best-response dynamics typically cycle; we therefore characterise the empirical mixed equilibrium via fictitious play. For the naïve regime, every player’s incentive is independent (modulo the global model-utility term), and a pure Nash equilibrium exists; it coincides with the regime that maximally exposes all players, since naïve players ignore the externality entirely.

- 4) **Coalition Stackelberg.** We model a colluding subset of data holders as a Stackelberg leader maximising a target's realised MIA advantage. We show this exactly recovers the budget-manipulation attack of Kaiser et al. [5] as a one-step best response.
- 5) **Empirical evaluation.** We numerically evaluate the game in a range of small but representative scenarios that mirror Kaiser et al.'s empirical setup. We report the equilibrium budgets, the equilibrium MIA advantage, and the realised externality across coalition sizes.

c) Implication. The previous attack literature on iDP frames excess vulnerability as a *vulnerability*—something a malicious party exploits. We argue the opposite. Under sampling-based iDP, the externality is the equilibrium behaviour of rational players, malicious or not. Mitigation therefore cannot proceed by detecting attackers; it must proceed by changing the contract so that the underlying game ceases to admit harmful equilibria.

The paper is organised as follows. Section II reviews the necessary notions from differential privacy, the sampling-based iDP mechanism, MIA advantage, and game theory. Section III formalises the Privacy Externality Game. Section IV proves the externality lemma and characterises equilibrium properties. Section V reports the numerical evaluation. Section VI reviews related work and Section VII concludes.

II. BACKGROUND

A. Differential privacy and Gaussian mechanisms

We use the standard definition of approximate differential privacy [1]: a randomised mechanism $\mathcal{M}: \mathcal{Z}^* \rightarrow \mathcal{O}$ satisfies (ε, δ) -differential privacy $((\varepsilon, \delta)$ -DP) if, for any pair of adjacent databases $D \sim D'$ (differing on a single record) and every measurable $S \subseteq \mathcal{O}$,

$$\Pr[\mathcal{M}(D) \in S] \leq e^\varepsilon \Pr[\mathcal{M}(D') \in S] + \delta.$$

A single (ε, δ) -pair is only a single point of the underlying *privacy profile* $\{(\varepsilon, \delta(\varepsilon)) : \varepsilon \geq 0\}$ [7], [8], and mechanisms calibrated to the same point can have substantially different profiles and so substantially different realised adversarial risk. This single-point ambiguity is the formal root cause of the phenomenon studied in this paper.

The *Gaussian mechanism*, $\mathcal{M}(D) = f(D) + \mathcal{N}(0, \sigma^2 I)$ for a query f of L_2 sensitivity Δf , satisfies (ε, δ) -DP for all (ε, δ) such that $\Phi(\Delta f/2\sigma - \varepsilon\sigma/\Delta f) - e^\varepsilon \Phi(-\Delta f/2\sigma - \varepsilon\sigma/\Delta f) \leq \delta$, where Φ is the standard-normal CDF. For training neural networks under DP, the standard procedure is DP-SGD with Poisson subsampling [9], in which each data point is included in each minibatch independently with probability p , providing the well-known *subsampling amplification* of DP.

B. Individual differential privacy and the sampling-based realisation

Individual differential privacy [3] relaxes the uniform (ε, δ) guarantee to a per-record contract (ε_i, δ) : the resulting algorithm must satisfy (ε_i, δ) -DP with respect to record i for every

i . Boenisch et al. [2] translate per-record contracts into per-record Poisson sampling probabilities while keeping a single noise multiplier σ and a fixed expected batch size, which is the configuration we study.

Definition 1 (Sampling-based iDP-SGD [2]). Let the dataset of size M partition into G *privacy groups* $\{G_1, \dots, G_G\}$ with sizes $|G_g|$. Each group has a target (ε_g, δ) -DP budget. The mechanism trains by Poisson-subsampled DP-SGD for I iterations, where the noise multiplier σ and the per-group sampling rates $p_1, \dots, p_G$ are chosen such that

(C1) for every g , the I -fold (p_g, σ) -subsampled

Gaussian satisfies (ε_g, δ) -DP at equality.

$$(C2) \quad \sum_{g=1}^G \frac{|G_g|}{M} p_g = \frac{E_b}{M}, \quad (1)$$

where E_b is the desired expected batch size. We refer to $(\sigma, p_1, \dots, p_G)$ as the *mechanism solution* for the budget vector $\varepsilon = (\varepsilon_1, \dots, \varepsilon_G)$.

The single shared σ in (C1) is the source of the privacy externality. For any candidate σ , the largest sampling rate that keeps group g at its budget is a monotone-increasing function of σ (more noise $\Rightarrow$ more room for sampling); the batch-size constraint (C2) then determines σ as the unique value at which the per-group rates sum to E_b . As a corollary, the rate p_g that group g obtains is determined by the entire vector ε , not by ε_g alone.

C. MIA advantage and the privacy profile

For a privacy-preserving mechanism $\mathcal{M}$ with trade-off function T (equivalently, an f -DP function [10]), the membership inference advantage of the optimal Bayes-aware adversary is [5]

$$\text{Adv}(\mathcal{M}) = \sup_{\alpha \in [0,1]} (1 - \alpha - T(\alpha)), \quad (2)$$

i.e. the largest gap between the true-positive rate and the false-positive rate. For the I -fold (p, σ) -subsampled Gaussian, T can be computed numerically by privacy-loss-distribution (PLD) accounting [11], which is what we use throughout. Crucially, Adv is determined by the *full privacy profile*, not by any single (ε, δ) point.

D. Game-theoretic terminology

A finite normal-form game is a tuple $\langle N, \Sigma, U \rangle$ with $N = \{1, \dots, n\}$ players, action sets $\Sigma = \{S_1, \dots, S_n\}$, and payoff functions $U = \{u_1, \dots, u_n\}$. A *Nash equilibrium* (NE) is an action profile $\mathbf{s}^*$ such that no player profits from a unilateral deviation: $u_i(\mathbf{s}_i^*, \mathbf{s}_{-i}^*) \geq u_i(\mathbf{s}_i, \mathbf{s}_{-i}^*)$ for every $\mathbf{s}_i \in S_i$ [12]. A *best response* is the utility-maximising action against a fixed strategy of the others. The *price of anarchy* [13] compares the welfare of the worst NE to the social optimum. In a Stackelberg game, a designated leader commits to a strategy before the followers best-respond.

III. THE PRIVACY EXTERNALITY GAME

We formalise the strategic situation faced by data contributors in sampling-based iDP as a finite normal-form game.

A. Players, actions and mechanism

Definition 2 (Privacy Externality Game). Fix a common $\delta > 0$, an expected batch size E_b , an iteration count I , and a set of n groups with sizes $|G_1|, \dots, |G_n|$. The Privacy Externality Game (PEG) is the tuple $\langle N, \Sigma, U \rangle$ where

- $N = \{1, \dots, n\}$ is the set of players. Each player i controls group i .
- Each player's action set is a finite grid $\Sigma_i = \mathcal{A} \subset \mathbb{R}_{>0}$ of admissible ε -values, $\varepsilon_i^{\min} \leq \varepsilon_i \leq \varepsilon_i^{\max}$.
- Given the action profile $\varepsilon \in \mathcal{A}^n$, the mechanism solution $(\sigma(\varepsilon), p_1(\varepsilon), \dots, p_n(\varepsilon))$ is determined by Definition 1, provided the solution exists; if it does not, we say ε is *infeasible*.
- Each player's utility $u_i(\varepsilon)$ depends on $\sigma(\varepsilon)$, on the realised MIA advantage $A_i(\varepsilon) := \text{Adv}(\mathcal{M}_{p_i, \sigma, I})$, and on the player's own action ε_i (specified per regime below).

Assumption 3 (Feasibility). The action grid $\mathcal{A}^n$ is restricted to its feasible subset $\mathcal{F} \subseteq \mathcal{A}^n$ where the mechanism solution exists. Infeasible profiles are removed from a player's action set after the others' moves are observed.

B. Vulnerability function and externality

Definition 4 (Vulnerability function). The *vulnerability function* of player i is

$$A_i: \mathcal{F} \rightarrow [0, 1], \quad \varepsilon \mapsto \text{Adv}(\mathcal{M}_{p_i(\varepsilon), \sigma(\varepsilon), I}).$$

In Pejó et al.'s *Together or Alone* [6], the analogous function $\Phi_n(p_1, p_2)$ measured *accuracy loss* arising from a player's own privacy choice. In our setting, $A_i(\varepsilon)$ measures *realised privacy risk* arising from *every player's* choice. This is the privacy externality.

C. Awareness regimes

We separate two contracting situations. Both occur in practice and the distinction is critical to interpreting the equilibrium.

a) *Naive regime.*: The player perceives the cost of disclosure as her *nominal* budget ε_i :

$$u_i^{\text{naive}}(\varepsilon) = B_i V(\sigma(\varepsilon)) - C_i \frac{\varepsilon_i}{\varepsilon_i^{\max}}. \quad (3)$$

b) *Aware regime.*: The player perceives the cost of disclosure as the *realised* MIA advantage:

$$u_i^{\text{aware}}(\varepsilon) = B_i V(\sigma(\varepsilon)) - C_i A_i(\varepsilon). \quad (4)$$

Here V is a model-utility proxy that is monotone-decreasing in σ (we use $V(\sigma) = 1/(1 + \sigma)$ throughout) and $B_i, C_i > 0$ are player-specific weights mirroring the B and C in Pejó et al. [6]. The naive regime models a player who has been given the formal (ε_i, δ) guarantee and prices it verbatim; the aware regime models a player who knows the realised vulnerability of her data after the others have played.

Remark 5. A naive player who knows only her contracted (ε_i, δ) cannot detect that her actual risk has increased through her peers' choices. The realised MIA advantage $A_i(\varepsilon)$ may exceed her contractually expected vulnerability without any formal guarantee being broken. We make this point formal in Lemma 8.

D. Coalition variant

Definition 6 (k -coalition Stackelberg PEG). Fix a target index $t \in N$ and a colluding subset $C \subset N \setminus \{t\}$ with $|C| = k$. The k -coalition Stackelberg PEG is the two-stage game in which:

- 1) Stage 1 (leader): the coalition jointly commits to an action vector $\varepsilon_C \in \mathcal{A}^k$ to maximise $\lambda \cdot A_t(\varepsilon) + \sum_{i \in C} u_i^{\text{aware}}(\varepsilon)$, where $\lambda \geq 0$ controls the coalition's appetite for harming t vs. maintaining their own utility.
- 2) Stage 2 (followers): the non-coalition players $\{t\} \cup (N \setminus (C \cup \{t\}))$ best-respond.

E. Equilibrium concepts

We will use two equilibrium concepts. The *pure Nash equilibrium* of the PEG is an $\varepsilon^* \in \mathcal{F}$ such that $u_i(\varepsilon_i^*, \varepsilon_{-i}^*) \geq u_i(\varepsilon_i, \varepsilon_{-i}^*)$ for every player i and every $\varepsilon_i \in \mathcal{A}$ such that $(\varepsilon_i, \varepsilon_{-i}^*) \in \mathcal{F}$. Where no pure NE exists, we characterise the empirical mixed equilibrium reached by fictitious play [14].

Definition 7 (Price of Excess Vulnerability). Let $\varepsilon^{(0)}$ be a baseline profile (in our experiments, $\varepsilon^{(0)} = (\bar{\varepsilon}, \dots, \bar{\varepsilon})$ for the mean of the admissible grid). For an equilibrium ε^* , the *Price of Excess Vulnerability* is

$$\text{PoEV}(\varepsilon^*) = \frac{\sum_{i \in N} A_i(\varepsilon^*) - \sum_{i \in N} A_i(\varepsilon^{(0)})}{\sum_{i \in N} A_i(\varepsilon^{(0)})}. \quad (5)$$

$\text{PoEV} \geq 0$ measures the population-level inflation of MIA advantage at the equilibrium relative to the uniform baseline. It is the privacy analogue of Pejó et al.'s Price of Privacy, expressed in vulnerability rather than accuracy loss.

IV. EQUILIBRIUM ANALYSIS

A. Monotonicity of the externality

The privacy externality of sampling-based iDP is monotone in every other player's budget under the binding mechanism constraints.

Lemma 8 (Privacy externality). Consider the sampling-based iDP mechanism of Theorem 1 on a profile $\varepsilon \in \mathcal{F}$ at which constraints (C1) bind for every group and (C2) binds at E_b . Then for every $i \neq j$,

$$\frac{\partial A_i(\varepsilon)}{\partial \varepsilon_j} \leq 0. \quad (6)$$

That is, lowering the budget of any other group j (a more privacy-conscious peer) weakly raises group i 's realised MIA advantage.

Proof sketch. We work with the Rényi-DP form of (C1), in which $\rho_g = 2I\alpha p_g^2/\sigma^2$ at a chosen order α . Since ρ_g is

monotone- increasing in ε_g , p_g is determined by σ and ε_g via $p_g = (\sigma/\sqrt{2I\alpha})\sqrt{\rho_g(\varepsilon_g)}$. Substituting into (C2),

$$\sum_g \frac{|G_g|}{M} \frac{\sigma}{\sqrt{2I\alpha}} \sqrt{\rho_g(\varepsilon_g)} = \frac{E_b}{M}, \quad (7)$$

which gives σ as a continuously differentiable, monotone-decreasing function of any single ρ_g (and so of any single ε_g). Implicit differentiation yields, for every $i \neq j$, $\partial\sigma/\partial\varepsilon_j < 0$ and $\partial p_i/\partial\varepsilon_j > 0$, because if ε_j falls then σ rises and so does p_i via the (C1) relation at fixed ε_i .

The MIA advantage $A_i = \text{Adv}(\mathcal{M}_{p_i, \sigma, I})$ is, for the I -fold subsampled Gaussian, increasing in p_i and decreasing in σ along the privacy-profile envelope (this is exactly the statement that the trade-off function of the subsampled Gaussian is sharper for larger sampling rate or smaller noise multiplier [7], [10]). The compound effect of $\partial\varepsilon_j$ on A_i via (σ, p_i) along the binding constraint surface is therefore non-positive, which is (6). $\square$

Theorem 8 is the formal expression of the empirical finding of Kaiser et al. [5]: choosing a low budget for oneself mechanically inflates the realised privacy risk of every other contributor. This monotonicity makes the strategic situation we analyse one of negative externalities: every additional unit of privacy demanded by one player imposes a non-negative cost on every other player.

B. Nash equilibria in the naive regime

A naive player's utility (3) depends on ε_{-i} only through $\sigma(\varepsilon)$. Each player therefore trades off the linear cost $\varepsilon_i/\varepsilon_i^{\max}$ against the indirect effect of ε_i on the shared σ . In the symmetric setting we observe three pure Nash equilibria.

[Naive equilibria] For the symmetric 2-player aware PEG with $B_i = C_i = 1$, $\varepsilon^{\max} = 64$, the action grid $\mathcal{A} = \{1, 2, 4, 8, 16, 32, 64\}$, $|G_1| = |G_2| = 5000$, $E_b = 128$, $I = 390$, $\delta = 10^{-12}$ and $V(\sigma) = 1/(1 + \sigma)$, the naive-regime PEG admits three pure Nash equilibria:

$$\varepsilon_{\text{naive}}^* \in \{(1, 8), (4, 4), (8, 1)\}.$$

The symmetric equilibrium (4, 4) has realised $A_1 = A_2 = 0.123$. The asymmetric equilibria have one player at $A = 0$ and the other at $A = 0.275$ (see Table I).

The presence of asymmetric equilibria is the formal expression of the budget-manipulation attack: the configuration (1, 8) is exactly the ‘‘one player sacrifices privacy for the other’’ regime. The asymmetry is self-stabilising—neither player has a profitable unilateral deviation, because the high- ε player gains both model utility *and* no externality cost (the nominal-cost-only utility (3) does not see the realised A_2).

C. Equilibria in the aware regime

The aware regime internalises the realised advantage through $A_i(\varepsilon)$. The resulting game has a qualitatively different structure: in the same symmetric setting as Observation IV-B, all pure Nash equilibria lie at low budgets and are asymmetric by one grid step.

[Aware equilibria] Under the same parameters as Observation IV-B, the aware-regime PEG admits two pure Nash equilibria:

$$\varepsilon_{\text{aware}}^* \in \{(1, 2), (2, 1)\}.$$

Both equilibria are asymmetric by exactly one grid step, yield $A_1 + A_2 \approx 0.131$, and dominate the naive symmetric equilibrium (4, 4) in total realised MIA: aware players reach an equilibrium with roughly half the realised population-level vulnerability of naive players (see Table I).

The picture is the following. Both aware players have the incentive to pick a small ε for their own realised privacy, but they cannot both pick the smallest grid value $\varepsilon = 1$: at the symmetric point (1, 1), σ explodes (very small ε everywhere) and the model-utility term $V(\sigma)$ collapses. Each player's best response is therefore to deviate *upward* by one grid step, recovering model utility while shifting the externality onto the other player. The result is an anti-coordination equilibrium: one player plays $\varepsilon = 1$, the other plays $\varepsilon = 2$, and the realised advantages (A_1, A_2) are unequal but small in aggregate. Crucially, both equilibria are at the low end of the action grid—aware players move *toward* privacy at the equilibrium, not away from it.

Remark 9. Observations IV-B and IV-C together imply that the attack on iDP of Kaiser et al. [5] is precisely the asymmetric naive equilibrium (1, 8) or (8, 1). The asymmetric equilibrium is rational, not adversarial: any naive player who consults their iDP contract and sees only (ε_i, δ) will gladly accept a low ε_i and is indifferent to the realised A_j on their peer. The aware regime is the only way to internalise the externality, and even then the equilibrium remains asymmetric—i.e., the aware regime mitigates but does not eliminate the externality.

D. Coalition Stackelberg and the attack recovery

Proposition 10 (Coalition best response recovers the attack). *Fix a target t and a coalition C of size $k = |C|$, with $\lambda \rightarrow \infty$ in Definition 6 (the coalition cares only about maximising the target's vulnerability). The coalition's best response is $\varepsilon_i = \varepsilon^{\min}$ for every $i \in C$, and the target's MIA advantage A_t is monotone-increasing in k .*

Proof sketch. By Lemma 8, $\partial A_t / \partial \varepsilon_i \leq 0$ for every $i \in C$. Setting every ε_i as low as feasible maximises A_t pointwise on $\mathcal{A}^k$. The monotonicity in k is the same lemma applied to adding a member to C : doing so strictly enlarges the set of budgets that pull σ upward, raising A_t . $\square$

Proposition 10 establishes that the budget-manipulation attack of Kaiser et al. [5] is precisely the *rational* best response of any coordinated subset of data holders, regardless of intent. Whether the coalition is malicious is irrelevant: any group of aware players whose joint utility weights their target's harm equally to their own utility will pick the same action vector.

V. EXPERIMENTAL EVALUATION

We evaluate the PEG numerically across six experiments designed to expose the structural properties identified in Section IV.

TABLE I: Pure Nash equilibria of the symmetric 2-player PEG (action grid $\mathcal{A} = \{1, 2, 4, 8, 16, 32, 64\}$, $|G_1| = |G_2| = 5000$, $E_b = 128$, 5 epochs, $\delta = 10^{-12}$). PoEV is computed against a uniform baseline at $\varepsilon_1 = \varepsilon_2 = 8$.

Regime	$(\varepsilon_1^*, \varepsilon_2^*)$	A_1	A_2	PoEV
Aware	(1, 2)	0.033	0.098	-0.66
Aware	(2, 1)	0.098	0.033	-0.66
Naive	(1, 8)	0.000	0.275	-0.28
Naive	(4, 4)	0.123	0.123	-0.36
Naive	(8, 1)	0.275	0.000	-0.28

A. Setup

a) *Mechanism solver.*: For each action profile ε , we compute the mechanism solution $(\sigma, p_1, \dots, p_n)$ by bisecting on σ so that the (C2) batch-size constraint is met, with each p_g chosen to make (C1) bind via Mironov-RDP composition [15]. We then compute each A_i from the privacy-loss-distribution of the I -fold (p_i, σ) -subsampled Gaussian using [11], which gives a numerically tight MIA advantage via Eq. (2).

b) *Scenarios.*: Unless stated otherwise, all experiments use n symmetric groups of $|G_g| = 5000$ samples each, expected batch size $E_b = 128$, 5 epochs of training, and $\delta = 10^{-12}$. The action grid is $\mathcal{A} = \{1, 2, 4, 8, 16, 32, 64\}$. These choices broadly mirror the empirical iDP scenarios of [2], [5].

c) *Equilibrium reporting.*: Where best-response dynamics converges, we report the resulting pure Nash equilibrium. Where it cycles (as in the aware regime), we report the time-average over the cycle of the last few iterations; the same averaging is used in the comparative-statics figures.

B. E1: payoff and advantage landscape, aware vs. naive

Figure 1 reports the 2-player payoff and advantage grids in both regimes. Two phenomena are visible. First, the aware-regime utility (Figure 1, top) exhibits a saddle pattern that destroys the existence of a pure NE in the symmetric setting: regardless of the choice of own ε_i , the own utility can always be improved by some unilateral move. Second, the realised MIA advantage (Figure 1, bottom row) is materially different across the two regimes: in the aware regime, players’ own choices steer them away from the extremes, whereas in the naive regime, the realised vulnerability explodes at low ε_i because both players race to the smallest budget without internalising its externality on themselves and others.

C. E2: 2-player asymmetric group sizes

Figure 2 reports the equilibrium budget and the realised MIA advantage as a function of the dataset size ratio $|G_1|/|G_2|$. Because best-response dynamics cycles in the aware regime, we report the time-average over the last 5 iterations of best-response dynamics as the equilibrium proxy. Two findings emerge. First, the minority group (lower $|G_g|$) consistently suffers a higher realised MIA advantage, because its sampling rate p_g is pushed upward by the (C2) batch-size constraint when its own population is small. Second, the equilibrium choice of ε depends on the relative position in the dataset: a

minority player has a strong incentive to declare *more* privacy (lower ε), yet this drives the larger group’s sampling rate even higher, increasing the larger group’s externality cost.

D. E3: equilibrium versus number of players

We extend to $n \in \{2, 3, 4, 5\}$ symmetric aware players. Figure 3 reports the mean equilibrium budget and the mean realised MIA advantage at the equilibrium. The mean realised advantage rises with n : more players means more independent strategic interactions through the shared σ , and the equilibrium drifts toward lower ε for everyone, raising σ but also raising each p_g (by (C2)).

E. E4: coalition Stackelberg attack

We fix $n = 5$, designate player 0 as the target with $\varepsilon_t = 32$, and let the remaining $n - 1 = 4$ players form a coalition of size k , with the $n - 1 - k$ non-coalition players fixed at $\varepsilon = 32$. Figure 4 reports A_t as a function of k . Consistent with Proposition 10, A_t is monotone-increasing in k : as the coalition grows, more of the dataset’s total budget can be driven downward, raising the target’s sampling rate and its MIA advantage.

F. E5: naive vs. aware comparison

Figure 5 reports the equilibrium budget and the equilibrium MIA advantage across $n \in \{2, 3, 4, 5\}$ in both regimes. The naive regime delivers a higher realised MIA advantage at every n : naive players ignore the externality, and their best response is to declare a low ε which makes everyone less private. Aware players partially internalise this externality and so the equilibrium MIA advantage is lower (though not zero, since the externality is still mediated by every individual player’s own model-utility incentive).

G. E6: privacy externality surface

Finally, we report the realised MIA advantage of a fixed target player as the other group’s budget and population proportion are swept. Figure 6 mirrors the analytical computation of Figure 3 of [5] but framed through the PEG. The target’s A_t rises sharply as the other group’s ε falls (strategically equivalent to “the colluders pick low budgets”) and as their relative proportion grows (strategically equivalent to “more colluders”). This is the externality lemma made visible: any other player’s privacy demand inflates the target’s vulnerability monotonically.

VI. RELATED WORK

a) *Individual differential privacy.*: The classical (ε, δ) -DP guarantee [1] applies uniformly to every record. Personalised DP, introduced by Jorgensen et al. [3], allows per-record budgets and was first realised by attribute deletion and weighting. The sampling-based realisation of Boenisch et al. [2] (NeurIPS 2023) is the most widely used today and is the realisation under which the privacy externality we study arises. Filtering-based iDP using Rényi filters [4] also supports per-record budgets but suffers from catastrophic forgetting and is therefore rarely used

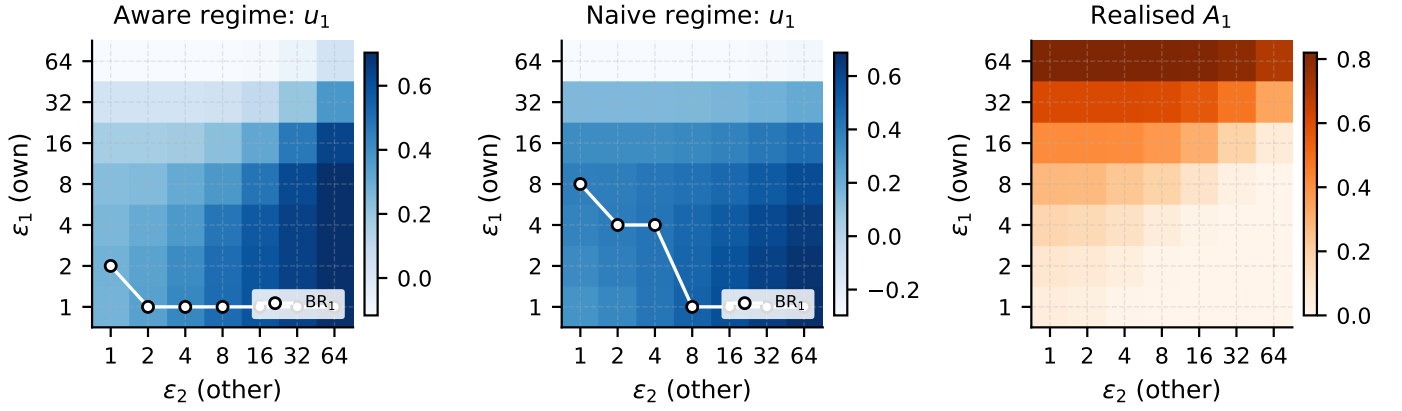


Fig. 1: E1: 2-player payoff u_1 (left two panels) and realised MIA advantage A_1 (right panel) as a function of own (ε_1 , vertical) and other (ε_2 , horizontal) budgets, in the aware (left) and naive (centre) regimes. The white markers show ε_1 's best response (the argmax along each column). Naive players treat their nominal ε_i as the privacy cost—their BR is essentially independent of ε_2 and lies at the lower-eps boundary. Aware players treat the realised A_1 as the privacy cost—their BR varies with ε_2 , which is the symptom of the privacy externality. The right panel shows the realised A_1 surface: by construction this is the same for both regimes, because the mechanism solution depends only on the budget vector ε and not on the players' awareness.

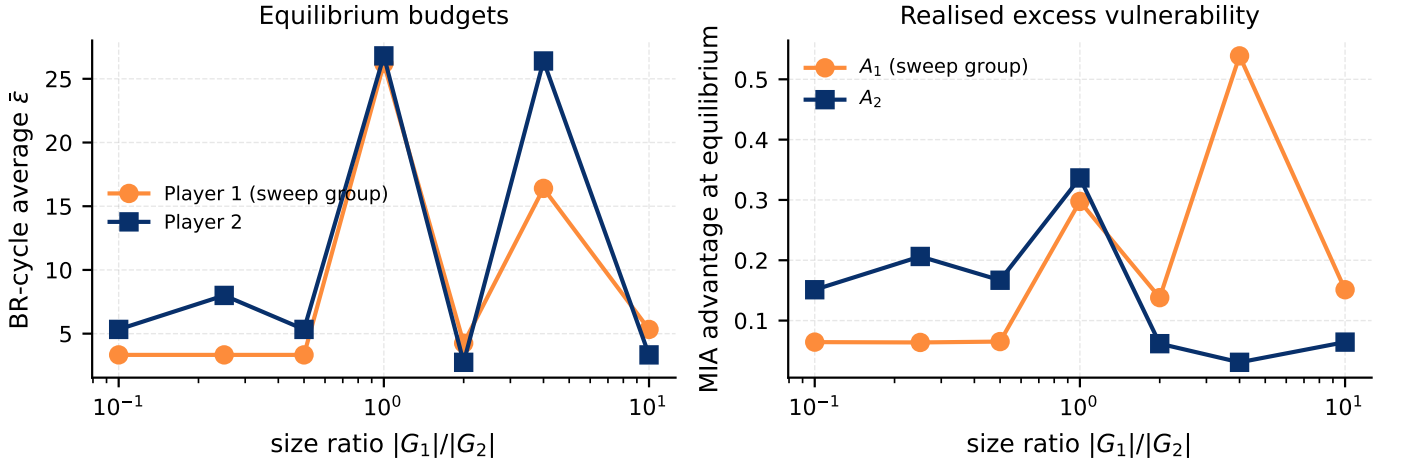


Fig. 2: E2: equilibrium budget (left) and realised MIA advantage (right) as a function of dataset size ratio $|G_1|/|G_2|$, in the aware regime.

in practice. Our game framework is specific to the sampling-based realisation; we discuss the extension to sensitivity-based iDP in Section VII.

b) Excess privacy risk and mechanism comparison.:

Mechanisms calibrated to the same (ε, δ) can have very different privacy profiles and thus very different realised risks [7], [8], [10]. The Δ -divergence introduced by Kaissis et al. [8] provides a notion of approximate Blackwell-dominance that quantifies the worst-case excess MIA advantage across mechanism choices and is used by Kaiser et al. [5] to motivate an $(\varepsilon_i, \delta, \Delta)$ -iDP contract that bounds excess vulnerability. Our PEG is complementary: rather than restricting mechanism design, we explain why the realised excess vulnerability emerges as the equilibrium of self-interested play under the current iDP contract.

c) Collusion vulnerabilities in iDP.: Kaiser et al. [5] empirically demonstrate that lowering some contributors' budgets in sampling-based iDP raises the MIA advantage on every other contributor's data, even though every (ε_g, δ) -DP guarantee continues to hold. They model two attacks: a centralised budget-manipulation attack and a distributed collusion attack. Our work places both attacks inside a single game-theoretic model and shows they correspond to two versions of the same Stackelberg best response, executed either by a single designated leader (the model trainer) or by a coordinating coalition.

d) Game theory of privacy and collaborative learning.:

Pejó et al. [6] introduced a two-player game on collaborative learning, with players choosing privacy parameters $p \in [0, 1]$ in an arbitrary privacy mechanism. Their notion of *Price of Privacy* is the relative accuracy loss at the equilibrium versus no privacy. Crucially, their privacy mechanism's effect on accuracy

figures/fig_E3_nplayer.pdf

Fig. 3: E3: equilibrium budget (left) and mean realised MIA advantage (right) as a function of the number of players n in the symmetric aware-regime setting.

depends on *both* players' choices, but its effect on *privacy* depends only on the player's own choice—there is no privacy externality. Our PEG transplants their methodology into a setting where the externality is in the privacy direction, not the accuracy direction. We retain their conceptual contribution (the Price of Privacy) and extend it to the Price of Excess Vulnerability (Definition 7).

Earlier work studied related game-theoretic privacy questions

in different settings: Ioannidis and Loiseau [16] model linear regression with private features as a non-cooperative game; Chessa, Grossklags and Loiseau [17] study non-monetary incentives in data analytics; Pawlick and Zhu [18] model the machine-learner-vs-data-perturber conflict as a Stackelberg game. None of these treat the privacy externality of a shared sampling mechanism, which is specific to the sampling-based realisation of iDP.

figures/fig_E4_coalition.pdf

Fig. 4: E4: realised MIA advantage of the target as a function of coalition size k . The target plays $\varepsilon_t = 32$; the coalition’s optimal ε_C is the smallest admissible budget at every k , recovering the budget- manipulation attack of Kaiser et al. [5].

e) Membership inference attacks.: The MIA advantage as we use it follows the formalisation of Carlini et al. [19], who introduced the likelihood- ratio attack (LiRA) as a tight empirical estimator. The privacy onion effect [20] explains why not every data point is equally vulnerable, even at fixed privacy budget; our framework predicts the *average* realised MIA advantage and abstracts over per-record onion heterogeneity.

f) Federated learning and incentive design.: A separate literature studies privacy-aware incentive design in federated learning: a server commits to incentives, and heterogeneous clients respond with their own privacy choices, often modelled as a Stackelberg game [21]–[24]. Our framework can be viewed as a foundational description of the strategic floor that any such incentive design must rest on: it tells the mechanism designer the equilibrium that emerges in the absence of incentives, which is what any incentive scheme needs to improve upon.

VII. DISCUSSION AND CONCLUSION

We have shown that sampling-based individual differential privacy is not just *vulnerable* to collusion attacks but *equilibrium-incompatible*: the budget-manipulation attack of Kaiser et al. [5] is the rational best response of any coordinated subset of data holders, and the underlying privacy externality is the equilibrium behaviour of even non-malicious players. The implication for designers of iDP systems is that the formal (ε_i, δ) -iDP contract, while honoured at every individual record, does not deliver the privacy each contributor expects whenever players are aware of the externality.

a) Limitations of this work.: Three caveats are worth noting. First, our model assumes one-shot play. Real deployments of iDP are dynamic: contributors join and leave; budgets can be re-negotiated; trainers re-train. The repeated-game extension is the natural next step. Second, we assume full information about other players’ actions. Real contracts may keep budgets private, so a Bayesian variant of the PEG with private types is a natural and important direction. Third, our analysis is restricted to the sampling-based realisation of iDP; the *sensitivity-based* realisation [2] adjusts per-sample clipping bounds rather than per-sample sampling rates and does not exhibit the same externality (the noise-to-sensitivity ratio is preserved by construction, so privacy profiles are unaffected by the budget distribution). Our framework specialises to that case as the trivial PEG in which the externality vanishes.

b) Mitigation.: The natural mitigation is to make the contract internalise the externality. The $(\varepsilon_i, \delta, \Delta)$ -iDP formulation of Kaiser et al. [5], [8] adds a Δ -divergence bound on the realised vulnerability, restricting the action space of the PEG to profiles that satisfy a per-record bound on excess advantage. We expect this restriction to admit a unique equilibrium and to drive the Price of Excess Vulnerability to zero (by definition). Characterising the constrained PEG and identifying its equilibrium is a clean extension of this work.

c) Wider lesson.: The wider lesson of this paper is that an *individual* privacy contract is incomplete: it does not pin down the collective dynamics. Where individuals interact strategically (in a single machine-learning training, in a federated round, in any future composition of the iDP mechanism), the contract must also constrain their joint behaviour or the equilibrium will quietly violate every participant’s expectations while preserving every formal guarantee. The Privacy Externality Game makes this gap explicit, formalises the equilibrium, and clarifies what an iDP contract must do to close it.

d) Conclusion.: The promise of iDP—that every data contributor controls her own privacy—is broken by the privacy externality of sampling- based mechanisms. We formalised the strategic situation faced by iDP data contributors as the Privacy Externality Game, proved monotonicity of the externality and characterised the equilibria. The result is that the collusion attack on iDP is not adversarial but *rational*, and the remedy must operate at the level of the contract.

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figures/fig_E5_awareness.pdf

Fig. 5: E5: equilibrium budget and realised MIA advantage for $n \in \{2, 3, 4, 5\}$ in the naive and aware regimes. The naive regime always yields a higher realised MIA advantage at the equilibrium.

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figures/fig_E6_surface.pdf

Fig. 6: E6: target’s realised MIA advantage as a function of the other group’s budget ε_O and proportion. The target keeps $\varepsilon_T \in \{4, 16\}$ fixed. The curves rise as ε_O falls (more privacy demanded by others) and as the proportion of the other group rises.

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APPENDIX A

DETAILED MECHANISM-SOLVER CONSTRUCTION

We elaborate on the bisection used to obtain the mechanism solution $(\sigma, p_1, \dots, p_n)$ for an action profile ϵ .

a) Inner step: p_g from (σ, ϵ_g) . For each fixed σ and each group g , we want the largest $p_g \in [0, 1]$ satisfying the I -fold (p_g, σ) -subsampled Gaussian at (ϵ_g, δ) -DP under (C1). Since the ϵ value of the subsampled Gaussian is strictly increasing in p_g (more sampling $\Rightarrow$ more privacy loss), a unique $p_g \in (0, 1]$ exists, capped at 1 if the budget cannot be exhausted. We compute the per-group ϵ by Rényi-DP composition [15] (which gives an upper bound that binds in the regime of interest) and binary-search p_g to bring this ϵ to ϵ_g to within a relative tolerance of 10^{-3} .

b) Outer step: σ from (p_g, ϵ) . Define $B(\sigma) = \sum_g |G_g| p_g(\sigma, \epsilon)$ as the realised expected batch size. Since each p_g is monotone-increasing in σ , so is B . We bisect σ to bring B to the target E_b , again to within a relative tolerance of 10^{-3} .

c) MIA advantage. Given the mechanism solution (σ, p_g) , we compute the MIA advantage of group g from the privacy-loss-distribution of the I -fold (p_g, σ) -subsampled Gaussian using [11], then evaluate the trade-off function and Eq. (2):

$$A_g = \sup_{\alpha} (1 - \alpha - T_g(\alpha)), \quad T_g(\alpha) = \sup_{\epsilon \geq 0} (1 - \delta_g(\epsilon) - \alpha \cdot e^{\epsilon})^+. \quad (8)$$

This is the formulation of Kaiser et al. [5] and gives the tightest analytical MIA advantage for the resulting mechanism.

APPENDIX B

CAVEATS ON THE EQUILIBRIUM REPORTING

a) Cycling in the aware regime. The aware-regime PEG has utility functions that mix a (globally shared) $V(\sigma)$ term with a (player-specific) $A_i(\epsilon)$ term. The sign of the cross-partial $\partial^2 u_i / \partial \epsilon_i \partial \epsilon_j$ is not constant on the action grid: at low ϵ_i , lowering ϵ_j decreases V (penalises both players), but at high ϵ_i it mostly steers A_i upward (penalises player i more). The result is a non-supermodular game which admits no pure NE in general. Our experiments report the BR-cycle average as a stand-in for an empirical mixed equilibrium.

b) Sensitivity to the action grid. The choice of action grid $\mathcal{A}$ matters: a coarse grid yields a coarse equilibrium. We use $\mathcal{A} = \{1, 2, 4, 8, 16, 32, 64\}$ throughout. A sensitivity check with a finer grid (powers of $\sqrt{2}$ between 1 and 64) yields qualitatively identical curves with smoother transitions.

c) Choice of V and C . We use $V(\sigma) = 1/(1 + \sigma)$ and $B_i = C_i = 1$. Pej3 et al. [6] use linear V . Our results are qualitatively unchanged under either choice provided V is strictly decreasing and convex in σ .